



C-SiS

CENTRE FOR SCIENCE IN SOCIETY

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

KOCHI - 682 022, KERALA, INDIA



No: C-SiS/132/Internship/2025-26/291

**CENTRE FOR SCIENCE IN SOCIETY
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY**

Kochi – 682022, Kerala, India

Certificate of Completion

This is to certify that

HARITHA PH

from

Mar Athanasius College Kothamangalam

has successfully completed the Science Communication Internship Programme, organized by the Centre for Science in Society (C-SiS) as part of the International Year of Quantum Science and Technology (IYQ) 2025.

The internship, held from 19/5/2025 to 30/5/2025 provided the intern with a comprehensive experience in exploring Quantum Science and honing skills in Science Communication, commemorating the 100th anniversary of Quantum Mechanics. The intern's contributions during the programme have been invaluable in bridging the gap between science and society, fostering awareness and understanding of Quantum Science among diverse audiences.

We extend our heartfelt congratulations and best wishes for their future endeavors.

Date of Issue:.....8/7/2025.....

Dr. P. Shaiju
Director, C-SiS
CUSAT



**INTERNATIONAL YEAR OF
Quantum Science
and Technology**

Internship Report

Quantum Mechanics @100

Fundamental Forces and Particles



Presented by:

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Guided by:

Faculty of Centre for Science in Society (C-SiS), CUSAT

Duration: May 2025-June 2025

Location: CUSAT, Kochi

This internship was part of the celebrations for the International Year of Quantum Science and Technology 2025.

Introduction

The year 2025 marks a century since the birth of quantum mechanics, a revolutionary branch of physics that reshaped our understanding of the universe. In celebration of this historic milestone, the Centre for Science in Society (C-SiS) at Cochin University of Science and Technology (CUSAT) organized the Science Communication Internship Programme as part of the International Year of Quantum Science and Technology (IYQ 2025). This initiative was conducted in collaboration with various departments of CUSAT, the LUCA Science Portal, Kerala Sastra Sahitya Parishad (KSSP), and other distinguished institutions and organizations.

The programme aimed to engage students across diverse disciplines in the exciting task of understanding and communicating quantum science. The core objective was to empower interns with the tools and training required to make complex scientific ideas accessible to the general public—bridging the gap between advanced research and public understanding.

As a selected intern, I had the opportunity to contribute to this initiative through a focused project on "Fundamental Forces and Particles"—a topic central to the Standard Model of particle physics and inherently connected to the quantum realm. My contribution involved the creation of a presentation, visual aids, and an exhibition concept aimed at effectively communicating this complex subject in an engaging and simplified manner.

This report details the learnings, processes, and creative approaches undertaken during the internship, culminating in a proposed public exhibition that showcases the beauty and power of quantum science through the lens of fundamental forces and particles.

Objective

The primary objectives of this internship project were:

To bridge the gap between scientific research and public understanding By interpreting complex topics in quantum physics—particularly fundamental forces and particles—into accessible and engaging content.

To contribute to the celebration of the International Year of Quantum Science and Technology (IYQ 2025) Through creative science communication efforts that highlight the historical significance and future relevance of quantum mechanics.

To conceptualize a public science exhibition That can be used in academic institutions or science outreach events, focusing on interactive and visually impactful displays on the Standard Model and quantum phenomena.

To develop personal skills in science communication Including storytelling, visual presentation, content simplification, and audience engagement techniques.

Timeline of Quantum Mechanics

The development of the Standard Model of particle physics is a remarkable tale of curiosity, theory, and experimental triumph. In 1928, British physicist Paul Dirac formulated an equation that not only united quantum mechanics and special relativity but also predicted the existence of antimatter, later confirmed with the discovery of the positron.

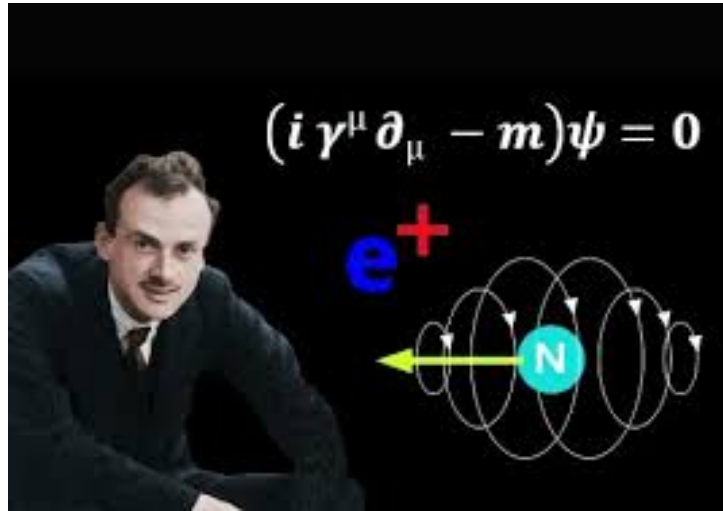


Figure 1: 1928—Paul Dirac introduced an equation predicting antimatter (positron).

Fast forward to 1954, when Chen-Ning Yang and Robert Mills extended the concept of gauge symmetry to the strong nuclear force, laying the groundwork for quantum chromodynamics.



Figure 2: 1954—Yang Chen-Ning and Robert Mills extended gauge theory to explain the strong force.

In 1928, Paul Dirac predicted the existence of antimatter through his relativistic equation for the electron. This marked the beginning of a new chapter in theoretical physics...

In 1957, physicist Chien-Shiung Wu experimentally confirmed that the weak nuclear force violates parity conservation, a profound insight that changed our understanding of symmetry in nature.



Figure 3: 1957—Chien-Shiung Wu proved that the weak force does not conserve parity (mirror image rule).

The journey continued in 1961, when Sheldon Glashow proposed a unified model of the weak and electromagnetic forces—known as the electroweak theory.

The Nobel Prize in Physics 1979



**Sheldon Lee
Glashow**
Prize share: 1/3



Abdus Salam
Prize share: 1/3



Steven Weinberg
Prize share: 1/3

The Nobel Prize in Physics 1979 was awarded jointly to Sheldon Lee Glashow, Abdus Salam and Steven Weinberg *"for their contributions to the theory of the unified weak and electromagnetic interaction between elementary particles, including, inter alia, the prediction of the weak neutral current"*.

Photos: Copyright © The Nobel Foundation

Figure 4: 1961—Sheldon Glashow combined electromagnetic and weak forces into one electroweak theory.

That same year, Murray Gell-Mann and George Zweig independently proposed the existence of quarks, the fundamental constituents of protons and neutrons.

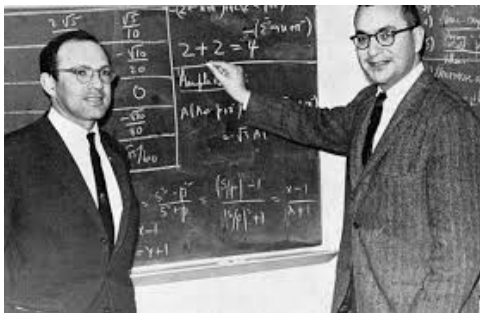


Figure 5: 1964—Murray Gell-Mann and George Zweig proposed the existence of quarks.

By 1964, Oscar Greenberg introduced the idea of color charge, explaining how quarks could obey the Pauli exclusion principle.



Figure 6: 1964—Oscar Greenberg introduced the idea of colour charge for quarks.

In 1967, Steven Weinberg and Abdus Salam refined the electroweak theory by introducing the Higgs mechanism, explaining how particles acquire mass.



Figure 7: 1967—Steven Weinberg and Abdus Salam added the Higgs mechanism to the electroweak theory.

In 1970, the GIM mechanism, developed by Glashow, Iliopoulos, and Maiani, predicted the existence of the charm quark, later confirmed experimentally.



Figure 8: 1970—Glashow, Iliopoulos, and Maiani predicted the charm quark (GIM mechanism).

A major breakthrough came in 1973, when Gross, Wilczek, and Politzer discovered asymptotic freedom, which explains how quarks behave more freely at high energies—a cornerstone of the strong force theory.



Figure 9: 1973—Gross, Wilczek, and Politzer discovered asymptotic freedom in strong interactions.

That same year, CERN experiments observed weak neutral currents, mediated by the Z boson.

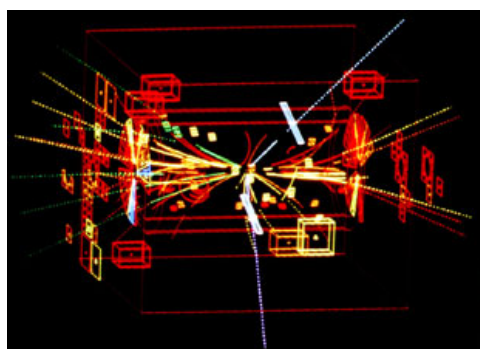


Figure 10: 1973—CERN experiments observed weak neutral currents (Z boson exchange).

The term "Standard Model" was formally introduced in 1975, capturing the growing theoretical framework. That year also saw the identification of the tau lepton by Martin Perl.

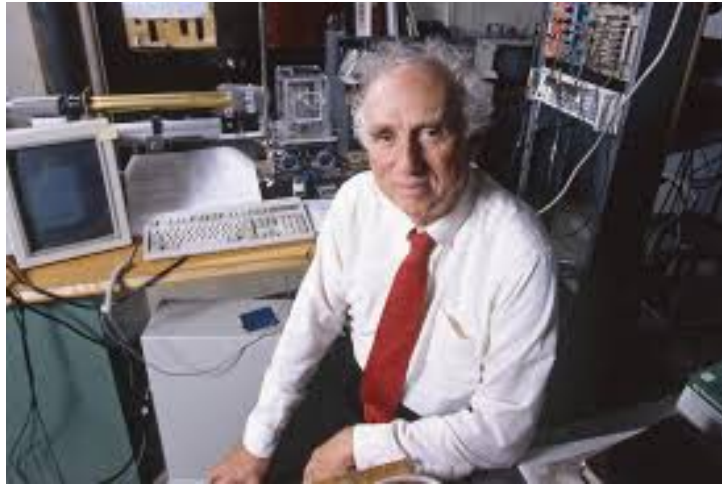


Figure 11: 1976—Martin Perl discovered the tau lepton.).

The discoveries kept coming: in 1977, Leon Lederman's team found the bottom quark; in 1983, CERN confirmed the existence of the W and Z bosons, central to the weak interaction.

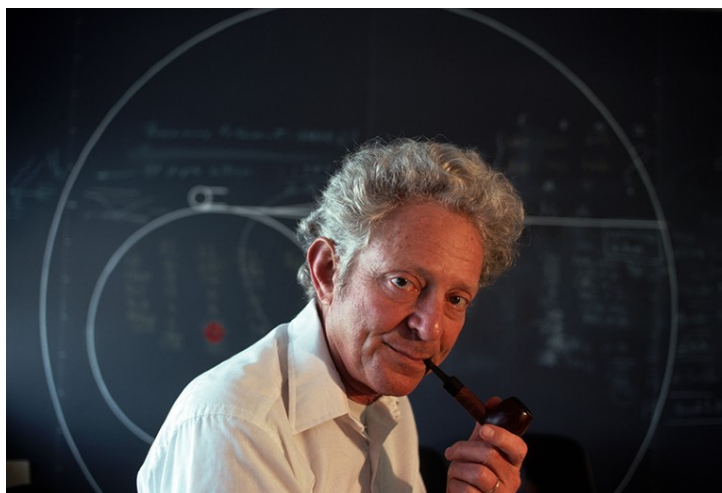


Figure 12: 1977—Leon Lederman's team discovered the bottom quark. 1983—W and Z bosons were discovered at CERN .).

The final pieces of the quark puzzle were placed in 1995, with the discovery of the top quark at Fermilab, and in 2000, when the elusive tau neutrino was confirmed. This incredible chapter culminated in 2012, when scientists at CERN's Large Hadron Collider

discovered the Higgs boson, validating the last missing component of the Standard Model and earning the nickname “the God particle.”

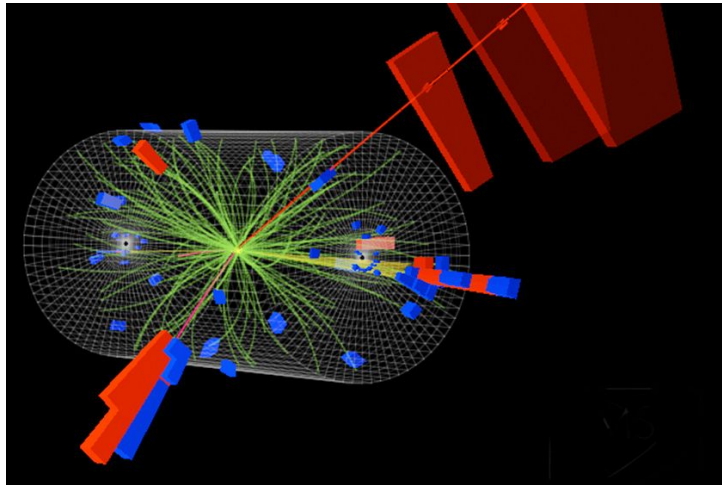


Figure 13: 2012—The Higgs boson was discovered at CERN.

This timeline is more than a series of discoveries—it is a testimony to decades of theoretical insight, experimental ingenuity, and international collaboration in the quest to understand the universe at its most fundamental level.

Fundamental Forces and Particles

The universe is governed by four fundamental forces:

- **Gravitational Force** – Weakest but with infinite range.
- **Electromagnetic Force** – Acts between charged particles.
- **Weak Nuclear Force** – Responsible for radioactive decay.
- **Strong Nuclear Force** – Binds quarks inside protons and neutrons.

These forces are carried by *force carriers*:

- Photon (EM)
- Gluon (Strong)
- W and Z bosons (Weak)
- Graviton (Theoretical, not yet detected)

The Standard Model classifies all known elementary particles: quarks, leptons, and bosons. It is our best framework for understanding matter and interactions.

The Standard Model

The Standard Model of particle physics is a well-established theory that describes the fundamental particles of the universe and the forces through which they interact—except gravity. It includes:

Fermions (matter particles): quarks and leptons

Bosons (force carriers): photon (electromagnetism), W/Z bosons (weak force), gluons (strong force), and the Higgs boson (mass generation)

The model successfully explains a wide range of phenomena, from the behavior of atoms to high-energy particle collisions in accelerators.

Importance: It predicts the existence of particles (like the Higgs boson) that were later discovered experimentally.

It forms the foundation of modern particle physics, guiding research at major facilities like CERN.

Despite its success, it leaves open questions—such as the nature of dark matter, gravity, and neutrino masses—suggesting it's part of a larger, deeper theory yet to be discovered.

Standard Model of Elementary Particles

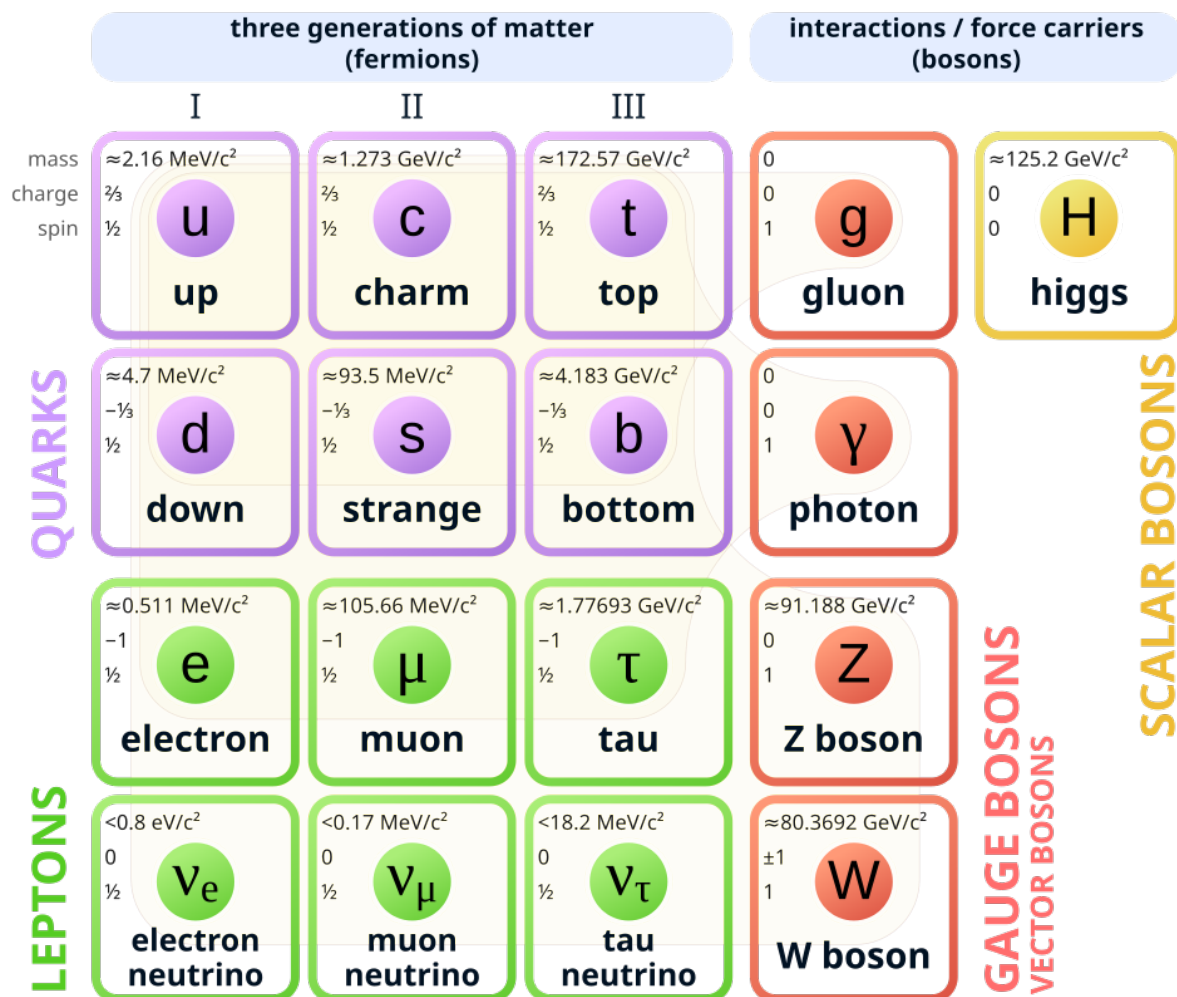


Figure 14: Standard Model Of Elementary Particles

Table 1: Distinction between Fermions and Bosons

Property	Fermions	Bosons
Spin	Half-integer (e.g., $1/2$, $3/2$)	Integer (e.g., 0, 1, 2)
Quantum Rule	Obey Pauli exclusion principle	Do not obey exclusion principle
Role in Universe	Make up matter (e.g., electrons, quarks)	Mediate forces (e.g., photons, gluons)
Wavefunction Symmetry	Antisymmetric	Symmetric
Statistics	Fermi-Dirac statistics	Bose-Einstein statistics
Examples	Electrons, protons, neutrons	Photons, W/Z bosons, Higgs boson

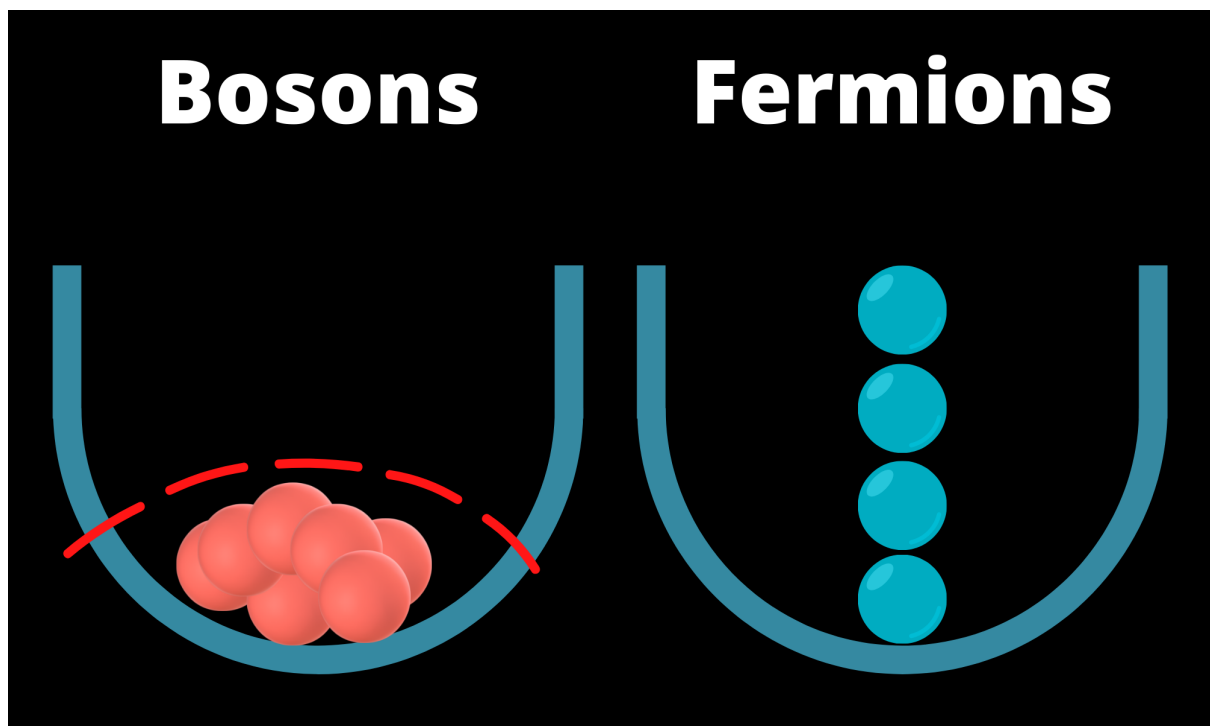


Figure 15: Bosons and Fermions

Fermions – The Building Blocks of Matter

Fermions are matter particles that obey the Pauli exclusion principle and follow Fermi–Dirac statistics. They possess **half-integer spin** (e.g. 1/2) and constitute the substance of the observable universe.

Types of Fermions

- **Leptons**
 - Electron, Muon, Tau
 - Corresponding neutrinos: ν_e , ν_μ , ν_τ
- **Quarks** (combine to form protons and neutrons)
 - Up, Down
 - Charm, Strange
 - Top, Bottom

Generations of Fermions

1. **First Generation** (common in everyday matter)
Electron, Up quark, Down quark, Electron neutrino
2. **Second Generation** (heavier counterparts)
Muon, Charm quark, Strange quark, Muon neutrino
3. **Third Generation** (heaviest, short-lived)
Tau, Top quark, Bottom quark, Tau neutrino

Higher generations are unstable and appear only in high-energy environments such as modern particle accelerators.

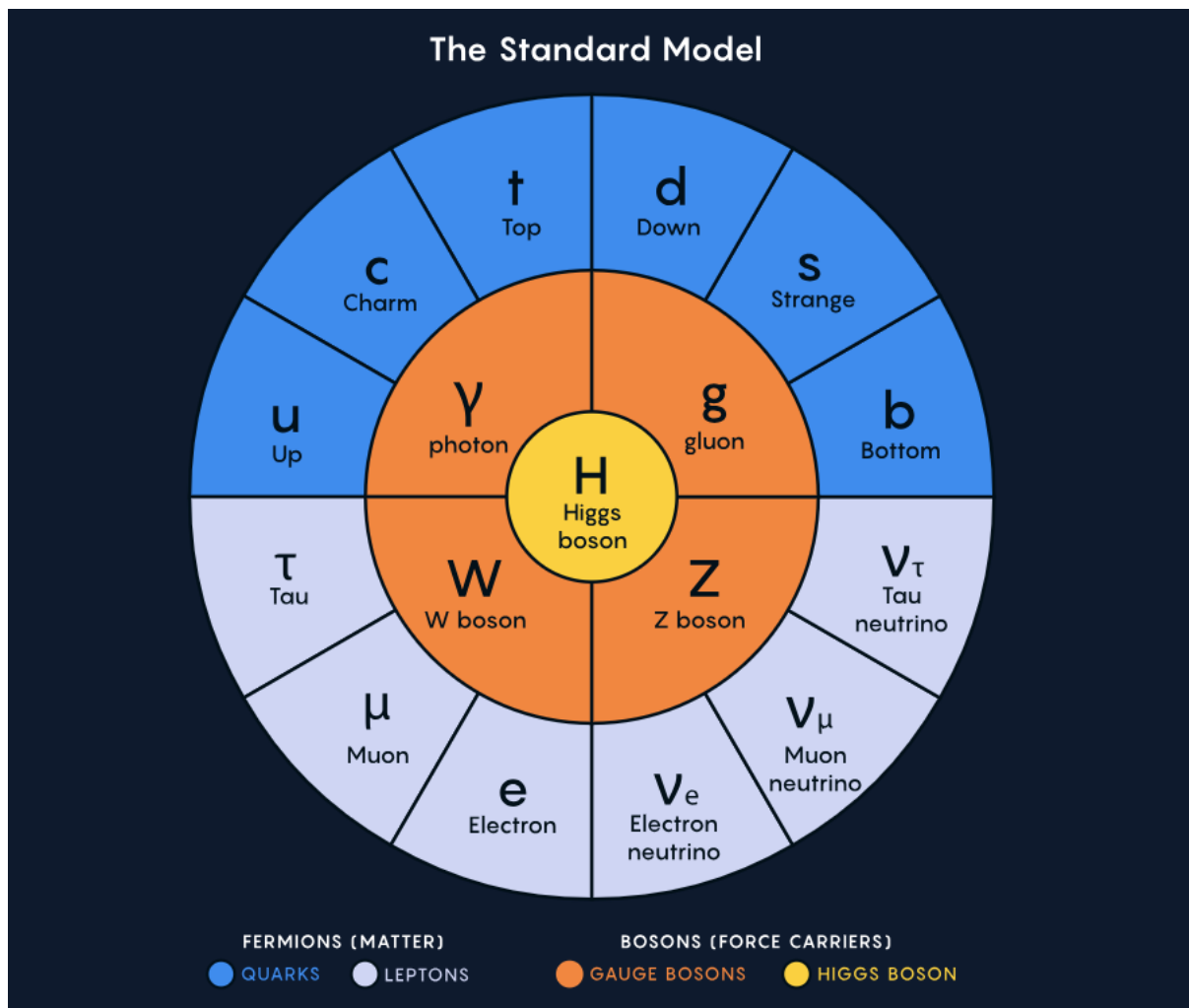


Figure 16: The Standard Model

Bosons – The Force Carriers

Bosons are particles that mediate the fundamental forces of nature. They obey Bose–Einstein statistics and have **integer spin** (e.g. 0, 1). Unlike fermions, multiple bosons can occupy the same quantum state.

Fundamental Bosons and the Forces They Mediate

- **Electromagnetic Force**
 - Mediated by the **photon** (γ)
 - Acts on charged particles
 - Governs electricity, magnetism, light, and chemical bonding

- **Strong Nuclear Force**

- Mediated by **gluons**
- Binds quarks together to form protons and neutrons
- Holds the atomic nucleus together

- **Weak Nuclear Force**

- Mediated by **$W^\pm$ and Z^0 bosons**
- Responsible for beta decay and nuclear fusion in stars

These bosons are essential to the Standard Model of particle physics. The Higgs boson, discovered in 2012, gives mass to other particles via the Higgs mechanism.

The Higgs Boson — The Universe's Invisible Tailor

The Role of the Higgs Boson

- A critical element of the **Standard Model** that explains how particles acquire mass.
- The **Higgs field** permeates all of space.
- Particles gain mass by interacting with the Higgs field — similar to moving through a viscous medium.
- The stronger the interaction, the more massive the particle.
- The **Higgs boson** is the quantum manifestation (excitation) of this field.
- Its discovery in **2012 at CERN's LHC** confirmed a major prediction of the Standard Model.

Without the Higgs mechanism, fundamental particles like electrons and quarks would be massless — and atoms, chemistry, and life as we know it wouldn't exist.

Limitations & Unsolved Mysteries

The Standard Model is a triumph of modern physics — but it is not the final story. Here are some of its major limitations:

- **No Gravity:** The Standard Model does not include gravity, described by Einstein’s theory of General Relativity. It works brilliantly at the quantum level but breaks down when applied to massive objects like black holes or the Big Bang.
- **Dark Matter & Dark Energy:** Together, these form about 95% of the universe’s total mass-energy content. Yet, they remain completely unexplained within the Standard Model.
- **Three Generations of Particles:** Quarks and leptons come in three nearly identical copies (generations) with increasing mass. The reason for this structure is still unknown.
- **Neutrino Oscillations:** Neutrinos can change from one type (flavor) to another as they travel, implying they have mass — contradicting the original Standard Model assumptions.
- **No Unified Framework:** The dream of unifying all fundamental forces, including gravity, under a single theoretical framework — often called the “Theory of Everything” — is still out of reach.

“The universe: 5% understood, 95% mystery. We’re basically still at Chapter 1.”

Beyond the Standard Model

- **Supersymmetry, Extra Dimensions, and String Theory:** These are theoretical extensions of the Standard Model. Supersymmetry (SUSY) proposes superpartners for all particles, potentially solving several open questions. Extra dimensions and string theory aim to unify quantum mechanics and gravity.
- **Future Experiments:**
 - **DUNE (Deep Underground Neutrino Experiment):** A next-gen neu-

trino project exploring neutrino mass hierarchy, CP violation, and proton decay.

- **FCC (Future Circular Collider):** A proposed 100 km collider to probe high-energy physics beyond LHC, searching for SUSY, dark matter, and more.
- **Grand Unified Theory (GUT):** Efforts continue to unify electromagnetic, weak, and strong forces under a single theoretical framework. The ultimate goal is a Theory of Everything including gravity.

Standard Model Exhibition Plan

What I Plan to Do

I aim to design a **chronological walkthrough** of the Standard Model's development, starting from the Dirac Equation in 1928 and moving decade by decade up to the Higgs Boson discovery in 2012 and future possibilities beyond.

Timeline Wall or Panel

- **Chronologically arranged pictures and discoveries** of key scientists:
 - Paul Dirac – predicted antimatter (1928)
 - Chien-Shiung Wu – confirmed parity violation
 - Glashow, Weinberg, Salam – developed electroweak theory
 - CERN team – discovered Higgs boson in 2012
- Each era will include photos, quotes, and story snippets.

Particle Models

- **3D or paper models** representing:
 - Electrons, quarks, neutrinos, gluons, Higgs boson
- Use **colored beads or clay** to show quark colors and force carriers.

Force Zones

- Interactive posters or charts of the **four fundamental forces**:
 1. Electromagnetic
 2. Strong Nuclear
 3. Weak Nuclear

4. Gravitational

- Include real-world examples and fun facts for each.

Interactive Corner

- **Build-an-Atom Activity:** Visitors assemble model atoms from parts.
- **Guess-the-Particle Quiz:** Clues and characteristics guide the guess.

Future Beyond the Standard Model

- Panels introducing advanced theories:
 - **Supersymmetry, String Theory, Extra Dimensions**
- Mention of ongoing and future experiments:
 - **DUNE:** Focused on neutrino physics
 - **FCC:** Aiming to build the next big collider

Reflection

This internship was a valuable experience that helped me grow both personally and academically. I learned how to simplify complex physics concepts without losing their scientific accuracy, which strengthened my interest in **science communication**. I also got a chance to explore new **technical skills**, especially in **LaTeX** and **Python**, which I had never used much before.

The sessions gave me an introduction to **various topics** in modern physics, especially **quantum physics**, in a way that felt both engaging and accessible. I was able to interact with **professors and students** from different backgrounds, which made the learning process more collaborative and interesting.

Through the activities conducted by **C-SiS**, I understood how hands-on and fun-based methods—like **games and interactive exhibits**—can help children understand physics more effectively. Seeing how simple tools can make science more approachable was inspiring.

Overall, this internship deepened my passion for **particle physics** and **science education**. It gave me insight into how science can be communicated creatively, and I'm grateful for the opportunity to be part of it.

Conclusion

This internship at **C-SiS, CUSAT** has been a truly eye-opening experience. It helped me see physics from a new perspective—not just as equations and theories, but as a subject that can be made engaging and meaningful for people of all ages. I learned to simplify complex topics while maintaining scientific accuracy, and I gained hands-on experience in communication, design, and interactive science education.

Most importantly, this experience deepened my curiosity and commitment toward science outreach. I now feel more confident and inspired to take part in future efforts that make science more accessible and enjoyable for everyone.

Acknowledgement

I sincerely thank the faculty members and coordinators of the **Centre for Science in Society (C-SiS)**, **CUSAT**, for providing this wonderful opportunity. Their constant guidance, encouragement, and approachable nature played a crucial role in shaping my understanding of science communication.

Throughout the internship, they patiently explained concepts, answered our questions, and motivated us to explore beyond our textbooks. Their efforts helped me gain confidence in presenting complex ideas in simpler forms. I truly appreciate the time and energy they devoted to making this internship an enriching experience.

This opportunity not only improved my communication and presentation skills but also inspired me to contribute actively to science education in the future. I am deeply grateful for the learning environment they created and for being part of this meaningful journey.

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